

11.0000

1

H

hydrogen

2.2

1312.0

-1 +1

1s¹

36.94

3

Li

lithium

0.98

520.2

+1

[He]2s¹

9.0122

4

Be

beryllium

1.57

899.5

+2

[He]2s²

22.990

11

Na

sodium

0.93

495.8

+1

[Ne]3s¹

24.305

12

Mg

magnesium

1.31

737.8

+2

[Ne]3s²

39.098

19

K

potassium

0.82

418.8

+1

[Ar]4s¹

40.078

20

Ca

calcium

1

589.8

+2

[Ar]4s²

44.956

21

Sc

scandium

1.36

633.1

+3

[Ar]3d¹4s²

47.867

22

Ti

titanium

1.54

658.8

+2 +3 +4

[Ar]3d²4s²

50.942

23

V

vanadium

1.63

650.9

+2 +3 +4 +5

[Ar]3d³4s²

51.996

24

Cr

chromium

1.66

652.9

+2 +3 +6

[Ar]3d⁵4s¹

54.938

25

Mn

manganese

1.55

717.3

+2 +3 +4 +6 +7

[Ar]3d⁵4s²

55.845

26

Fe

iron

1.83

762.5

+2 +3 +6

[Ar]3d⁶4s²

58.933

27

Co

cobalt

1.88

760.4

+2 +3

[Ar]3d⁷4s²

58.693

28

Ni

nickel

1.91

737.1

+2 +3

[Ar]3d⁸4s²

63.546

29

Cu

copper

1.9

745.5

+1 +2

[Ar]3d¹⁰4s¹

65.38

30

Zn

zinc

1.65

906.4

+2

[Ar]3d¹⁰4s²

69.723

31

Ga

gallium

1.81

578.8

+3

[Ar]3d¹⁰4s²4p¹

72.630

32

Ge

germanium

2.01

762.2

-4 +2 +4

[Ar]3d¹⁰4s²4p²

74.922

33

As

arsenic

2.18

944.5

-3 +3 +5

[Ar]3d¹⁰4s²4p³

78.971

34

Se

selenium

2.55

941.0

-2 +2 +4 +6

[Ar]3d¹⁰4s²4p⁴

79.904

35

Br

bromine

2.96

1139.9

-1 +1 +3 +5 +7

[Ar]3d¹⁰4s²4p⁵

83.798

36

Kr

krypton

—

1350.8

+2

[Ar]3d¹⁰4s²4p⁶

85.468

37

Rb

rubidium

0.82

403.0

+1

[Kr]5s¹

87.62

38

Sr

strontium

0.95

549.5

+2

[Kr]5s²

88.906

39

Y

yttrium

1.22

599.9

+3

[Kr]4d¹5s²

91.222

40

Zr

zirconium

1.33

640.1

+4

[Kr]4d²5s²

92.906

41

Nb

niobium

1.6

652.1

+3 +5

[Kr]4d⁴5s¹

95.95

42

Mo

molybdenum

2.16

684.3

+4 +6

[Kr]4d⁵5s¹

97

43

Tc

technetium

2.1

686.9

+4 +7

[Kr]4d⁵5s²

101.07

44

Ru

ruthenium

2.2

710.2

+2 +3 +4 +8

[Kr]4d⁷5s¹

102.91

45

Rh

rhodium

2.28

719.7

+1 +3

[Kr]4d⁸5s¹

106.42

46

Pd

palladium

2.2

804.4

+2 +4

[Kr]4d¹⁰

107.87

47

Ag

silver

1.93

731.0

+1

[Kr]4d¹⁰5s¹

112.41

48

Cd

cadmium

1.69

867.8

+2

[Kr]4d¹⁰5s²

114.82

49

In

indium

1.78

558.3

+1 +3

[Kr]4d¹⁰5s²5p¹

118.71

50

Sn

tin

1.96

708.6

-4 +2 +4

[Kr]4d¹⁰5s²5p²

121.76

51

Sb

antimony

2.05

830.6

-3 +3 +5

[Kr]4d¹⁰5s²5p³

127.60

52

Te

tellurium

2.1

869.3

-2 +2 +4 +6

[Kr]4d¹⁰5s²5p⁴

126.90

53

I

iodine

2.66

1008.4

-1 +1 +3 +5 +7

[Kr]4d¹⁰5s²5p⁵

131.29

54

Xe

xenon

2.6

1170.4

+2 +4 +6 +8

[Kr]4d¹⁰5s²5p⁶

132.91

55

Cs

caesium

0.79

375.7

+1

[Xe]6s¹

137.33

56

Ba

barium

0.89

502.8

+2

[Xe]6s²

178.49

72

Hf

hafnium

1.3

658.5

+4

[Xe]4f¹⁴5d²6s²

180.95

73

Ta

tantalum

1.5

728.4

+5

[Xe]4f¹⁴5d³6s²

183.84

74

W

tungsten

1.7

758.8

+4 +6

[Xe]4f¹⁴5d⁴6s²

186.21

75

Re

rhenium

1.9

755.8

+4 +7

[Xe]4f¹⁴5d⁵6s²

190.23

76

Os

osmium

2.2

814.2

+2 +3 +4 +8

[Xe]4f¹⁴5d⁶6s²

192.22

77

Ir

iridium

2.2

865.2

+3 +4

[Xe]4f¹⁴5d⁷6s²

195.08

78

Pt

platinum

2.2

864.4

+2 +4

[Xe]4f¹⁴5d⁹6s¹

196.97

79

Au

gold

2.4

890.1

+1 +3

[Xe]4f¹⁴5d¹⁰6s¹

200.59

80

Hg

mercury

1.9

1007.1

+1 +2

[Xe]4f¹⁴5d¹⁰6s²

204.38

81

Tl

thallium

1.8

589.4

+1 +3

[Xe]4f¹⁴5d¹⁰6s²6p¹

207.2

82

Pb

lead

1.8

715.6

+2 +4

[Xe]4f¹⁴5d¹⁰6s²6p²

208.98

83

Bi

bismuth

1.9

702.9

+3 +5

[Xe]4f¹⁴5d¹⁰6s²6p³

209

84

Po

polonium

2

812.2

-2 +2 +4

[Xe]4f¹⁴5d¹⁰6s²6p⁴

210

85

At

astatine

2.2

899.0

-1 +1

[Xe]4f¹⁴5d¹⁰6s²6p⁵

222

86

Rn

radon

—

1037.1

—

[Xe]4f¹⁴5d¹⁰6s²6p⁶

223

87

Fr

francium

0.7

393.0

+1

[Rn]7s¹

226

88

Ra

radium

0.9

509.3

+2

[Rn]7s²

138.91

57

La

lanthanum

1.1

538.1

+3

[Xe]5d¹6s²

140.12

58

Ce

cerium

1.12

534.4

+3 +4

[Xe]4f¹5d¹6s²

140.91

59

Pr

praseodymium

1.13

527.8

+3 +4

[Xe]4f³6s²

144.24

60

Nd

neodymium

1.14

533.1

+3

[Xe]4f⁴6s²

145

61

Pm

promethium

—

538.6

+3

[Xe]4f⁶6s²

150.36

62

Sm

samarium

1.17

544.5

+2 +3

[Xe]4f⁶6s²

151.96

63

Eu

europium

—

547.1

+2 +3

[Xe]4f⁷6s²

157.25

64

Gd

gadolinium

1.2

593.4

+3

[Xe]4f⁷5d¹6s²

158.93

65

Tb

terbium

—

565.8

+3 +4

[Xe]4f⁹6s²

162.50

66

Dy

dysprosium

1.22

573.0

+3

[Xe]4f¹⁰6s²

164.93

67

Ho

holmium

1.23

581.0

+3

[Xe]4f¹¹6s²

167.26

68

Er

erbium

1.24

589.3

+3

[Xe]4f¹²6s²

168.93

69

Tm

thulium

1.25

596.7

+2 +3

[Xe]4f¹³6s²

173.05

70

Yb

ytterbium

—

603.4

+2 +3

[Xe]4f¹⁴6s²

174.97

71

Lu

lutetium

1

523.5

+3

[Xe]4f¹⁴5d¹6s²

227

89

Ac

actinium

1.1

519.1

+3

[Rn]6d¹7s²

232.04

90

Th

thorium

1.3

608.5

+4

[Rn]6d²7s²

231.04

91

Pa

protactinium

1.5

568.3

+4 +5

[Rn]5f²6d¹7s²

238.03

92

U

uranium

1.7

597.6

+3 +4 +5 +6

[Rn]5f³6d¹7s²

237

93

Np

neptunium

1.3

604.5

+3 +4 +5 +6 +7

[Rn]5f⁴6d¹7s²

244

94

Pu

plutonium

1.3

581.4

+3 +4 +5 +6

[Rn]5f⁶7s²

243

95

Am

americium

—

576.4

+3

[Rn]5f⁷7s²

247

96

Cm

curium

—

578.2

+3

[Rn]5f⁷6d¹7s²

247

97

Bk

berkelium

—

598.0

+3 +4

[Rn]5f⁹7s²

251

98

Cf

californium

—

606.1

+3

[Rn]5f¹⁰7s²

252

99

Es

einsteinium

—

614.5

+3

[Rn]5f¹¹7s²

257

100

Fm

fermium

—

627.2

+3

[Rn]5f¹²7s²

258

101

Md

mendelevium

—

634.9

+2 +3

[Rn]5f¹³7s²

259

102

No

nobelium

—

639.3

+2 +3

[Rn]5f¹⁴7s²

266

103

Lr

lawrencium

—

478.6

+3

[Rn]5f¹⁴7s²7p¹

Radioactive sign

Atomic number

Atomic mass ([] = mass number of longest-lived isotope)

Symbol

Name

First ionisation energy (kJ/mol)

Electronegativity

Oxidation states (C: -4...+4 = every integer)

Electron configuration (ground state; predicted for Rf-Og)

92

238.03

U

uranium

597.6

1.7

+3 +4 +5 +6

[Rn]5f³6d¹7s²

5

10.81

B

boron

2.04

800.6

+3

[He]2s²2p¹

6

12.011

C

carbon

2.55

1086.5

-4...+4

[He]2s²2p²

7

14.007

N

nitrogen

3.04

1402.3

-3 +1 +2 +3 +4 +5

[He]2s²2p³

8

15.999

O

oxygen

3.44

1313.9

-2

[He]2s²2p⁴

9

18.998

F

fluorine

3.98

1681.0

-1

[He]2s²2p⁵

10

20.180

Ne

neon

—

2080.7

—

[He]2s²2p⁶

13

26.982

Al

aluminium

1.61

577.5

+3

[Ne]3s²3p¹

14

28.085

Si

silicon

1.9

786.5

-4 +4

[Ne]3s²3p²

15

30.974

P

phosphorus

2.19

1011.8

-3 +3 +5

[Ne]3s²3p³

16

32.06

S

sulfur

2.58

999.6

-2 +2 +4 +6

[Ne]3s²3p⁴

17

35.45

Cl

chlorine

3.16

1251.2

-1 +1 +3 +5 +7

[Ne]3s²3p⁵

18

39.95

Ar

argon

—

1520.6

—

[Ne]3s²3p⁶

31

69.723

Ga

gallium

1.81

578.8

+3

[Ar]3d¹⁰4s²4p¹

32

72.630

Ge

germanium

2.01

762.2

-4 +2 +4

[Ar]3d¹⁰4s²4p²

33

74.922

As

arsenic

2.18

944.5

-3 +3 +5

[Ar]3d¹⁰4s²4p³

34

78.971

Se

selenium

2.55

941.0

-2 +2 +4 +6

[Ar]3d¹⁰4s²4p⁴

35

79.904

Br

bromine

2.96

1139.9

-1 +1 +3 +5 +7

[Ar]3d¹⁰4s²4p⁵

36

83.798

Kr

krypton

—

1350.8

+2

[Ar]3d¹⁰4s²4p⁶

49

114.82

In

indium

1.78

558.3

+1 +3

[Kr]4d¹⁰5s²5p¹

50

118.71

Sn

tin

1.96

708.6

-4 +2 +4

[Kr]4d¹⁰5s²5p²

51

121.76

Sb

antimony

2.05

830.6

-3 +3 +5

[Kr]4d¹⁰5s²5p³

52

127.60

Te

tellurium

2.1

869.3

-2 +2 +4 +6

[Kr]4d¹⁰5s²5p⁴

53

126.90

I

iodine

2.66

1008.4

-1 +1 +3 +5 +7

[Kr]4d¹⁰5s²5p⁵

54

131.29

Xe

xenon

2.6

1170.4

+2 +4 +6 +8

[Kr]4d¹⁰5s²5p⁶

81

204.38

Tl

thallium

1.8

589.4

+1 +3

[Xe]4f¹⁴5d¹⁰6s²6p¹

82

207.2

Pb

lead

1.8

715.6

+2 +4

[Xe]4f¹⁴5d¹⁰6s²6p²

83

208.98

Bi

bismuth

1.9

702.9

+3 +5

[Xe]4f¹⁴5d¹⁰6s²6p³

84

209

Po

polonium

2

812.2

-2 +2 +4

[Xe]4f¹⁴5d¹⁰6s²6p⁴

85

210

At

astatine

2.2

899.0

-1 +1

[Xe]4f¹⁴5d¹⁰6s²6p⁵

86

222

Rn

radon

—

1037.1

—

[Xe]4f¹⁴5d¹⁰6s²6p⁶

113

286

Nh

nihonium

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p¹

114

290

Fl

flerovium

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p²

115

290

Mc

moscovium

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p³

116

293

Lv

livermorium

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p⁴

117

294

Ts

tennessine

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p⁵

118

294

Og

oganeson

—

—

—

[Rn]5f¹⁴6d¹⁰7s²7p⁶

Alkali metals

Alkaline-earth metals

Transition metals

Post-transition metals

Metalloids

Nonmetals

Halogens

Noble gases

Lanthanides

Actinides

Chemistry insufficiently characterised

Two classifications

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